Scenario A: Basic mixed workload

Input

Quantum = 3

Process	AT	BT
P1	0	10
P2	1	4
P3	2	6
P4	3	2

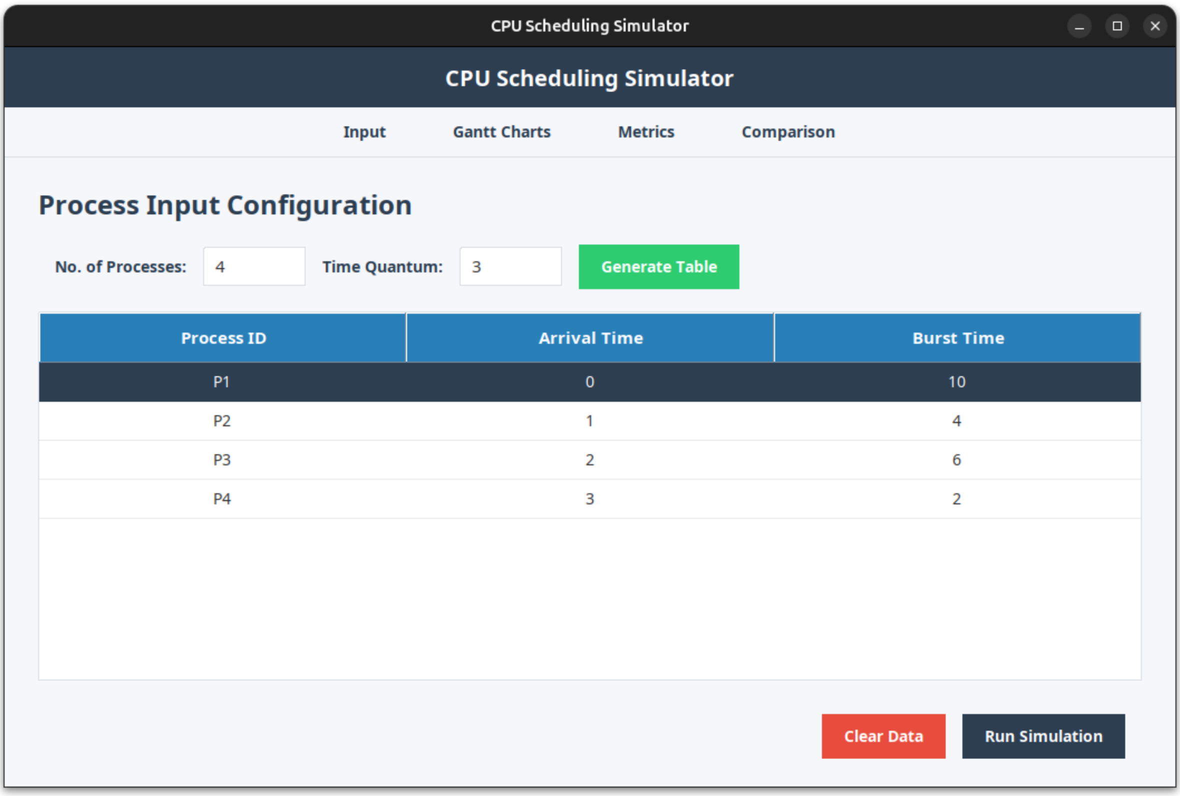
Analysis

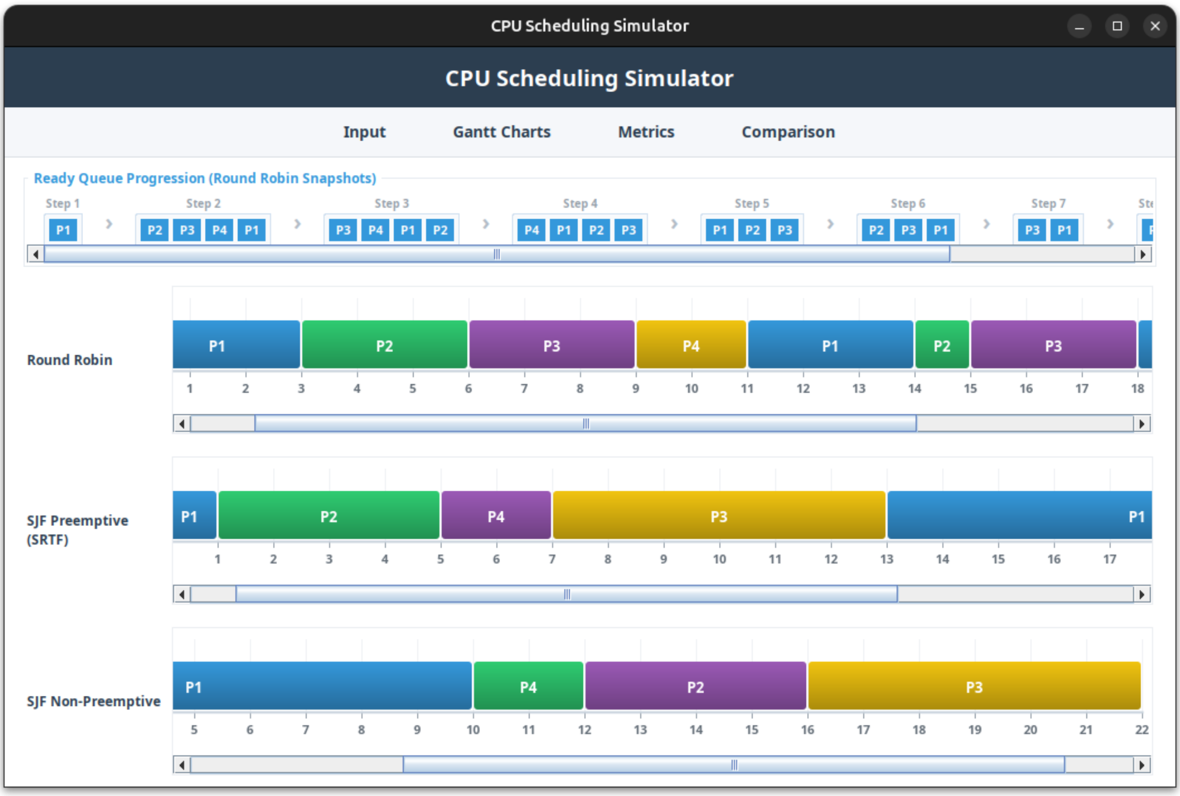
- **Which algorithm gave lower average waiting time?**
  SRTF gave the lowest average waiting time, followed by SJF, then Round Robin.
- **Which algorithm gave lower average response time?**
  SRTF achieved the best response time because it immediately switched to shorter jobs when they arrived.
- **Did Round Robin appear fairer across all processes?**
  Yes. Round Robin distributed CPU time fairly among all processes through cyclic execution.
- **Did SJF complete short jobs more efficiently?**
  Yes. SJF completed shorter processes earlier than Round Robin, but SRTF was even more efficient.
- **How did the chosen quantum affect Round Robin behavior?**
  The quantum caused frequent context switching and delayed completion of short processes compared to SJF and SRTF.
- **Which algorithm would you recommend for the tested workload, and why?**
  SRTF is recommended because it achieved the best overall efficiency in waiting time, turnaround time, and response time.

Conclusion

- SRTF performed best in all major performance metrics.
- Round Robin appeared to be the most balanced and fair algorithm.
- SJF was efficient but less adaptive than SRTF because it could not preempt running processes.

- The selected quantum in Round Robin increased waiting and turnaround time due to repeated time slicing and context switching.





CPU Scheduling Simulator

CPU Scheduling Simulator

Input | Gantt Charts | **Metrics** | Comparison

Round Robin Performance

Process ID	Waiting Time (WT)	Turnaround Time (TAT)	Response Time (RT)
P1	12	22	0
P2	10	14	2
P3	10	16	4
P4	6	8	6

Average WT: 9.50 | **Average TAT:** 15.00 | **Average RT:** 3.00

SJF Preemptive (SRTF) Performance

Process ID	Waiting Time (WT)	Turnaround Time (TAT)	Response Time (RT)
P1	12	22	0
P2	0	4	0
P3	5	11	5
P4	2	4	2

Average WT: 4.75 | **Average TAT:** 10.25 | **Average RT:** 1.75

SJF Non-Preemptive Performance

Process ID	Waiting Time (WT)	Turnaround Time (TAT)	Response Time (RT)
P1	0	10	0
P2	11	15	11
P3	14	20	14
P4	7	9	7

Average WT: 8.00 | **Average TAT:** 13.50 | **Average RT:** 8.00

CPU Scheduling Simulator

CPU Scheduling Simulator

Input | Gantt Charts | Metrics | **Comparison**

Comparative Scheduling Analysis

📊 Comparison Analysis

Waiting & Response Time:
- *SJF Preemptive (SRTF)* achieved the **lowest average waiting time**.
- *SJF Preemptive (SRTF)* provided the **fastest response** to new processes.

⚖ Fairness & Distribution:
- **Round Robin** is the most fair; waiting times are evenly distributed.

⏱ Quantum Effect (Q=3):
- Balanced quantum: Good trade-off between fairness and efficiency.

Final Result:
- Best Overall Waiting Time: **SJF Preemptive**